

Let's Investigate

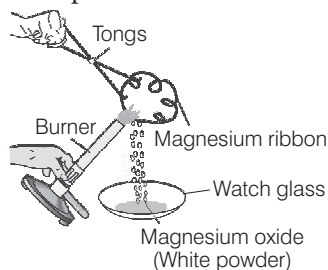
ACTIVITY 1 (Page 1)

Objective To study the combustion reaction (exothermic) by taking the example of burning of magnesium in air.

Materials Required Magnesium ribbon (approximately 2 cm long), sandpaper, a pair of tongs, burner and watch glass.

Procedure

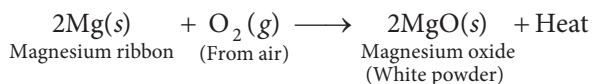
1. Take a piece of magnesium ribbon of nearly 2 cm length and rub it with a sandpaper.
2. Now, hold it with a pair of tongs and heat over the flame of a burner.
3. Collect the white powder formed in a watch glass.



Burning of a magnesium ribbon in air and collection of magnesium oxide in watch glass

Observation Magnesium ribbon burns with a dazzling or brilliant white flame and a white powder of magnesium oxide is formed.

Chemical Reaction



Conclusion Given reaction is a **combustion** (exothermic) reaction accompanied by the evolution of heat and light. Moreover, here oxidation of magnesium occurs.

Note When MgO is dissolved in water, it forms magnesium hydroxide which is basic in nature, i.e. it turns red litmus blue because of its basic nature.

Viva Questions

1. Combustion reaction is an example of physical change or chemical change.
2. What are the burning characteristics of magnesium ribbon?
3. Name the type of reaction between magnesium and oxygen.

4. Is the nature of magnesium oxide (MgO) acidic or basic?
5. Burning of magnesium is an exothermic or endothermic reaction?

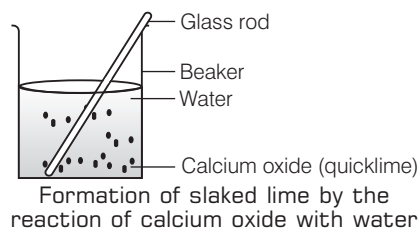
ACTIVITY 2 (Page 6)

Objective To study an exothermic combination reaction with the help of the reaction between calcium oxide and water.

Materials Required Quicklime, water, beakers (2), thermometer and glass rod.

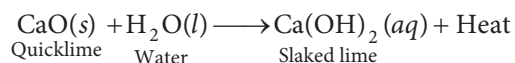
Procedure

1. Take a small amount (approximately 5 g) of quicklime (calcium oxide) in a beaker with a glass rod in it.
2. Take some water in another beaker and measure its temperature.
3. Now, add water slowly to the beaker containing quicklime.
4. Observe the changes and note the temperature change of the solution formed.



Observation The reaction is vigorous and the temperature rises as the reaction proceeds.

Chemical Reaction



Conclusion It is a **combination reaction** which is highly exothermic in nature.

Viva Questions

1. Write the chemical formula of quicklime.
2. Name the product formed when quicklime reacts with water.
3. Is there any change in temperature during the given reaction?

- A solution of a substance X is used for white washing. Name the substance X.
- Name the compound used to test the evolution of carbon dioxide gas.

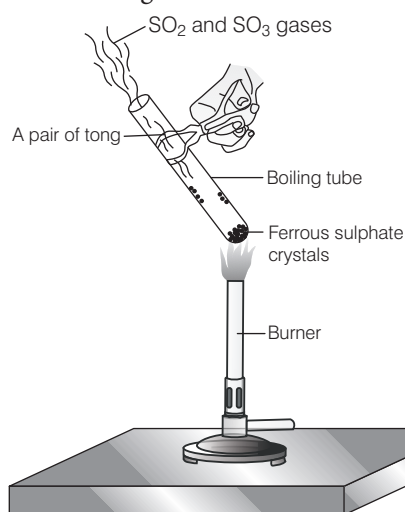
ACTIVITY 3 (Page 8)

Objective To study the decomposition of ferrous sulphate and lead nitrate.

Materials Required Ferrous sulphate crystals (2 g) lead nitrate crystals (2g), boiling tube, burner and a pair of tong.

Procedure

- Take some ferrous sulphate crystals in a dry boiling tube and heat it on the flame of a burner.
- Observe the changes.



Decomposition of ferrous sulphate crystals

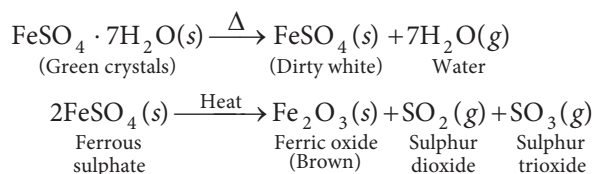
- Similarly, perform the same procedures for lead nitrate crystals and observe the changes.

Observation When green coloured ferrous sulphate crystals are heated, firstly a dirty white powder is formed (anhydrous salt), then on further heating, a brown residue is formed and characteristic odour of sulphur is observed.

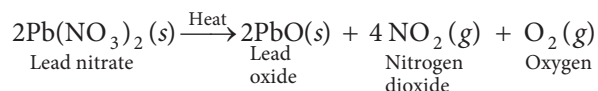
When lead nitrate is heated, emission of brown fumes is observed which is due to the formation of nitrogen dioxide (NO_2).

Chemical Reactions

Reactions involve in decomposition of ferrous sulphate.



Decomposition of lead nitrate,



Conclusion Decomposition of ferrous sulphate is an example of **dehydration and thermal decomposition** as the crystals first lose water molecules then break into the constituents on heating.

The characteristic odour of burning sulphur is due to SO_2 and SO_3 gases.

Decomposition of lead nitrate is also an example of thermal decomposition reaction.

Viva Questions

- What is the formula of crystalline ferrous sulphate?
- What is the other name of hydrated ferrous sulphate?
- Name the products formed when anhydrous ferrous sulphate is heated.
- What is the type of reaction when crystals of ferrous sulphate are heated strongly?
- What are the products formed in the decomposition of ferrous sulphate?

ACTIVITY 4 (Page 9)

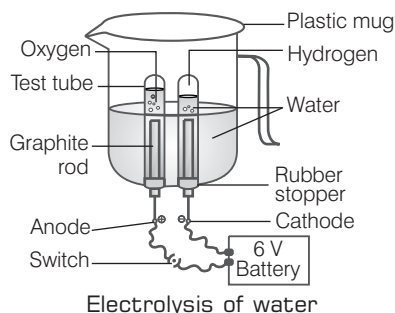
Objective To study the electrolysis of water.

Materials Required Plastic mug with 2 holes, a few drops of concentrated sulphuric acid, 2 graphite rods, 2 test tubes, water, wires and battery (6V), rubber stopper etc.

Procedure

- Take a plastic mug, drill two holes at the base, fit rubber stoppers in these holes and insert graphite rods (electrodes) in these rubber stoppers.
- Now connect these electrodes to a 6V battery with the help of wires.
- Fill the mug (upto half) with acidulated water. Remember that the electrodes should be immersed in water.

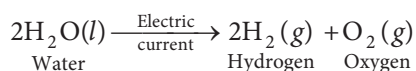
- Now take two test tubes filled with water and invert them over the two graphite electrodes. Test the gases evolved in test tubes.



Observation After some time, some gas bubbles arising in water at both the electrodes.

The gases formed at two electrodes go on collecting in the top parts of the inverted test tubes. In one test tube (at cathode), the volume of the gas (H_2) is double than the other (O_2) at anode.

Chemical Reaction



Conclusion This reaction is an example of decomposition reaction. As it takes place by passing electric current, so it is called **electrolysis**.

In water, H_2 and O_2 gases are present in ratio 2 : 1 by volume.

Note A few drops of acid are added to water that make it good conductor of electricity.

Viva Questions

- Which one is a chemical change, electrolysis of water or sodium chloride exposed in sunlight?
- In what ratio H_2 and O_2 are present in electrolytic decomposition of water?
- Why decomposition reaction of water is called electrolysis of water?
- What conclusion can you draw from electrolysis of water?
- Name the gas which burns with pop sound.

ACTIVITY 5 (Page 9)

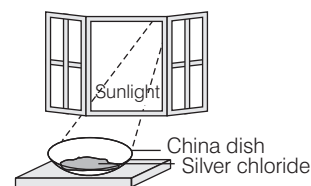
Objective To study the effect of sunlight on silver chloride.

Materials Required Silver chloride and China dish.

Procedure

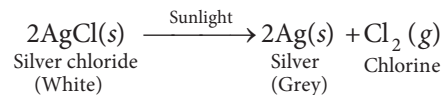
- Place a small amount of silver chloride (about 2 g) in a China dish. Observe its colour.

- Place this China dish in sunlight for some time and observe the colour change.



Observation After some time, the colour of the white silver chloride changes to grey.

Chemical Reaction



Conclusion This is a **decomposition reaction**. As it takes place in the presence of light, so it is also called **photolysis** or **photochemical decomposition**.

Viva Questions

- What change in colour is observed when white silver chloride (AgCl) is left exposed to sunlight?
- What type of chemical reaction is this?
- Name the products obtained when silver chloride is exposed to sunlight.
- Name two salts that are used in black and white photography.
- Write the decomposition reaction of silver bromide (AgBr).

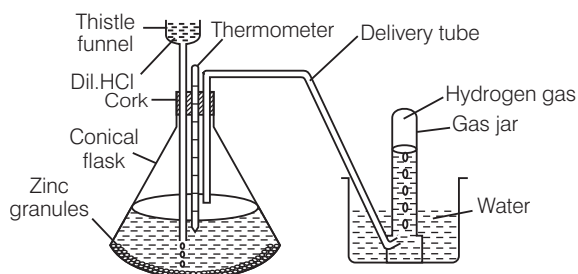
ACTIVITY 6 (Page 2, 10)

Objective To study the displacement reaction with the example of reaction between zinc and dilute hydrochloric acid.

Materials Required Zinc granules, dil.HCl, conical flask, thistle funnel, cork with three bores, delivery tube, gas jar and thermometer.

Procedure

- Take a few zinc granules in a conical flask and place a cork over it with a thistle funnel, thermometer and delivery tube.
- Now, add some dil. HCl solution in the conical flask with the help of thistle funnel.



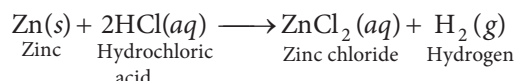
Formation of hydrogen gas by the action of dilute hydrochloric acid on zinc granules

- Note the temperature at the beginning and at the end of the reaction.
- Collect the gas produced in the gas jar and test it for identification.

Observation

- Bubbles of the gas are evolved. On testing, the gas is found to be hydrogen (burns with pop sound).
- The temperature rises during the reaction.

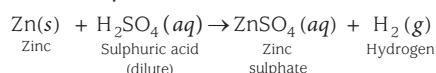
Chemical Reaction



Conclusion

- It is a displacement reaction. Zinc is present above hydrogen in the reactivity series of metals. Hence, Zn displaces it from HCl.
- The reaction is exothermic as it is accompanied by the evolution of heat.

Note This activity can also be performed by using dilute sulphuric acid instead of dilute hydrochloric acid. The reaction is as follows:



Viva Questions

- Which gas is produced when zinc granules react with dilute hydrochloric acid?
- Name the type of reaction when zinc reacts with dilute hydrochloric acid.
- Name the type of reaction in which heat is evolved.
- Why zinc granules are preferred over zinc strip for the reaction of Zn with HCl?
- How will you test for the presence of hydrogen gas?

ACTIVITY 7 (Page 10)

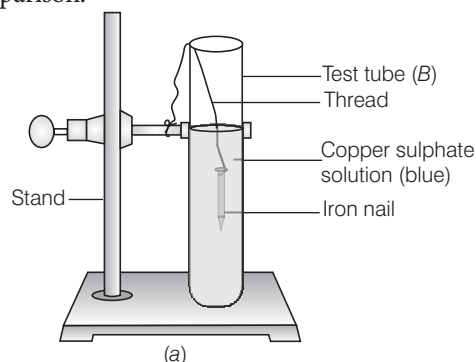
Objective To study the displacement reaction of copper by iron in copper sulphate solution.

Materials Required Iron nails (2), test tubes (2), copper sulphate solution and sand paper.

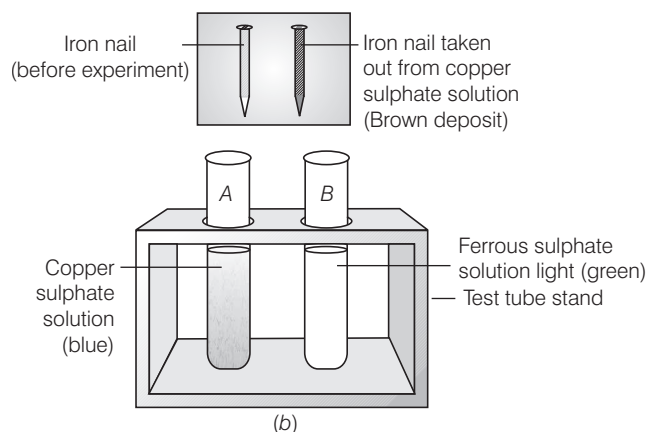
Procedure

- Take two iron nails and clean them by rubbing with sandpaper.
- Take two test tubes marked as A and B and put 10 mL CuSO_4 in both.

- Tie one iron nail with a thread and immerse it in test tube B, for about 20 min.
- Keep solution A as such and second iron nail aside for comparison.



(a) Iron nails dipped in copper sulphate solution in test tube B

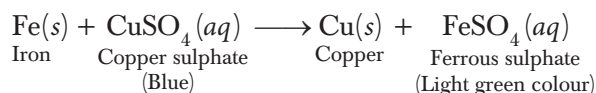


(b) Iron nails and copper sulphate solutions compared before and after the experiment

Observation The blue colour of copper sulphate solution in test tube B fades away, while the blue colour in test tube A remains as such.

The brown deposits appeared on the iron nail that is taken out from test tube B.

Chemical Reaction



Conclusion The more reactive element, iron displaces copper from copper sulphate solution. This is a displacement reaction.

Viva Questions

1. What changes in the colour of iron nails and copper sulphate solution do you observe after keeping the iron nails dipped in copper sulphate solution for about 20 min?
2. Why the blue colour of copper sulphate fades by adding iron nails in it?
3. What is the formula of crystalline copper sulphate?
4. Is it safe to store the solution of iron (II) sulphate in a copper vessel?
5. Name the type of reaction in this activity.

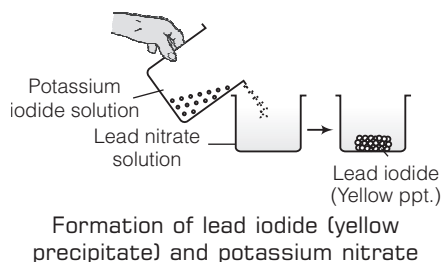
ACTIVITY 8 (Page 2/12)

Objective To study double displacement reaction by taking an example of reaction between lead nitrate and potassium iodide.

Materials Required Two beakers, lead nitrate, potassium iodide and water.

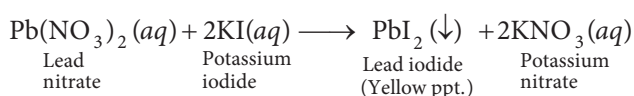
Procedure

1. Prepare the aqueous solutions of lead nitrate and potassium iodide in separate beakers.
2. Now, add potassium iodide solution slowly to the lead nitrate solution.



Observation A yellow precipitate of lead iodide is formed.

Chemical Reaction



Conclusion This reaction is an example of double displacement reaction as well as precipitation reaction.

Viva Questions

1. Name the type of reaction between lead nitrate and potassium iodide.
2. What is the colour of the precipitate formed?
3. Reaction between lead nitrate and potassium iodide is an example of two types of reactions. Name them.
4. What is the valency of lead (Pb) in lead nitrate?

5. Name the products formed in the reaction between lead nitrate and potassium iodide.

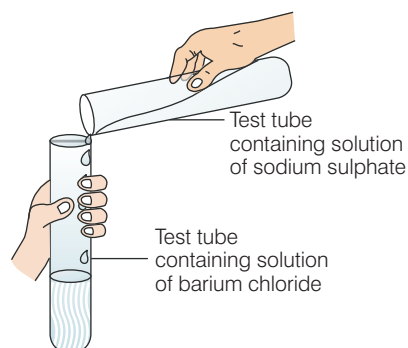
ACTIVITY 9 (Page 11)

Objective To study the double displacement reaction of barium chloride and sodium sulphate solution.

Materials Required Test tubes (2), 3 mL barium chloride solution and 3 mL sodium sulphate solution, stirrer rod and conical flask.

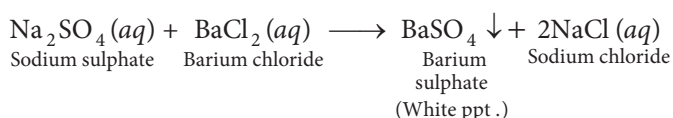
Procedure

1. Take 3 mL of sodium sulphate solution in one test tube and 3 mL of barium chloride solution in another test tube.
2. Then add these two solutions in conical flask and stirring it by using stirrer rod.



Observation A white substance, which is insoluble in water is formed. This insoluble substance is a precipitate of barium sulphate (BaSO_4).

Chemical Reaction



Conclusion The reaction is an example of double displacement reaction and precipitation reaction.

Viva Questions

1. Name the type of reaction in which two compounds exchange their ions to form two new compounds.
2. What is the insoluble substance formed in a reaction between barium chloride and sodium sulphate?
3. What is the colour of barium sulphate precipitate in a reaction?
4. State the physical conditions of reactants in which the reaction between barium chloride and sodium sulphate will not take place.
5. Write a chemical equation if ammonium sulphate is used in place of sodium sulphate in this activity.